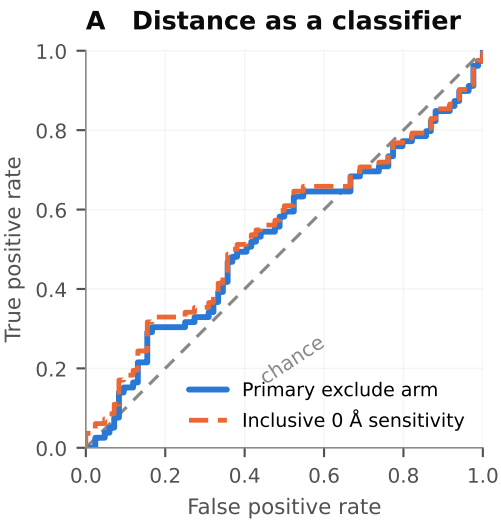


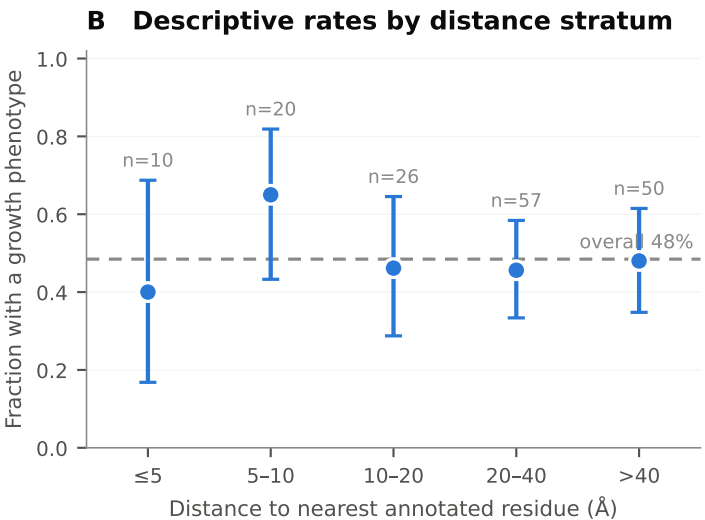
Exploratory calibration of AlphaFold distance to active and binding sites against yeast phosphomutant growth phenotypes

Yeast phospho-null mutants (Viéitez et al., Nat Biotechnol 2022) mapped onto AlphaFold models. All data public.

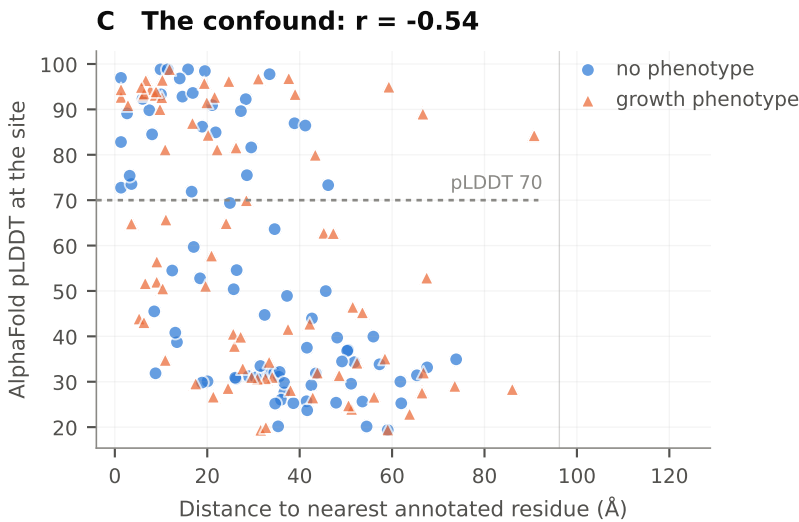
Primary AUC 0.527 95% CI [0.417, 0.632] (protein-clustered) · chance is 0.500
163 sites · 48 proteins · 79 with a growth phenotype
Inclusive 0 Å sensitivity: AUC 0.544 [0.436, 0.649] (n=166)



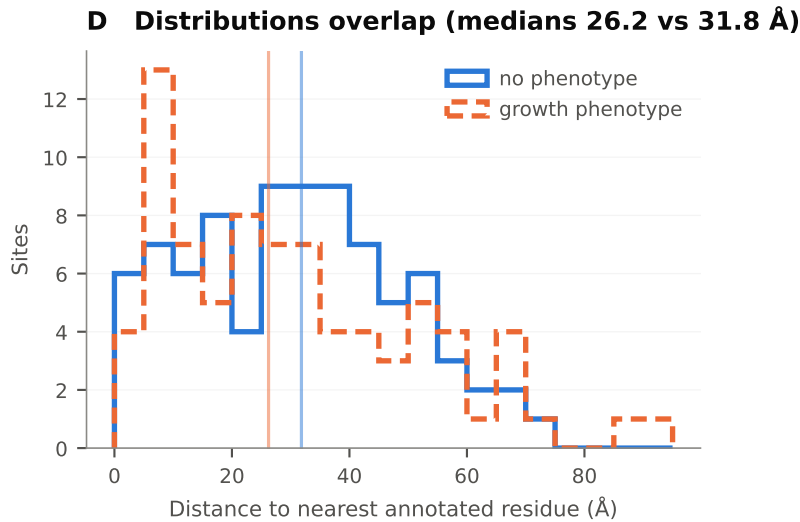
Shorter distance was oriented toward a positive outcome.
The protein-cluster interval is reported in the header.



Bins and Wilson intervals are descriptive at the substitution level;
the analysis was not powered around a universal cutoff.



Longer distances tend to occur at lower-confidence sites;
measurement confidence and biological context are entangled.



The observed distributions overlap substantially.
Vertical lines mark the medians.

497 point-mutant records / 116 genes → 474 with raw profiles → 427 sequence/QC-eligible records
→ 163 primary substitutions in 48 proteins after excluding 3 annotation-coincident sites; the named inclusive sensitivity contains 166 substitutions.

Cluster-robust logistic regression: OR 0.77 [0.27, 2.15] per 10-fold increase in distance + 1 Å.
Within 5 Å: 10 substitutions, 40.0% positive; beyond 5 Å: 49.0% positive. These are descriptive rates, not a threshold test.

Post-result SIFT comparator: AUC 0.606 [0.522, 0.690] on 152 substitutions. This is not independent validation. All analyses are exploratory; the interval around the distance AUC remains compatible with effects in either direction.
Public source data only. The growth outcome describes alanine-mutant phenotypes, not direct phosphorylation dependence. · Kyle Nguyen · 29 July 2026